School of Informatics & Cybersecurity - Assessment Submission Coversheet

Student Information

Student Number:	B00168689
Student Name:	Róisín Aylward

Module Information

Module Title:	Security Architecture
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Assessment Information

Assessment Title:	Designing and Implementing a Proof of Concept Cloud Solution based on a clients needs

Declaration

☒ I confirm that I have read and understood the University's Academic Integrity Policy.

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☒ I declare that this assessment submission is my own work.

☒ I confirm that I have read and understood the AI traffic light system used as part of this assessment brief and I have adhered to the level of AI usage allowed for this assessment.

Student signature:	Róisín Aylward
Date:	04/04/2026

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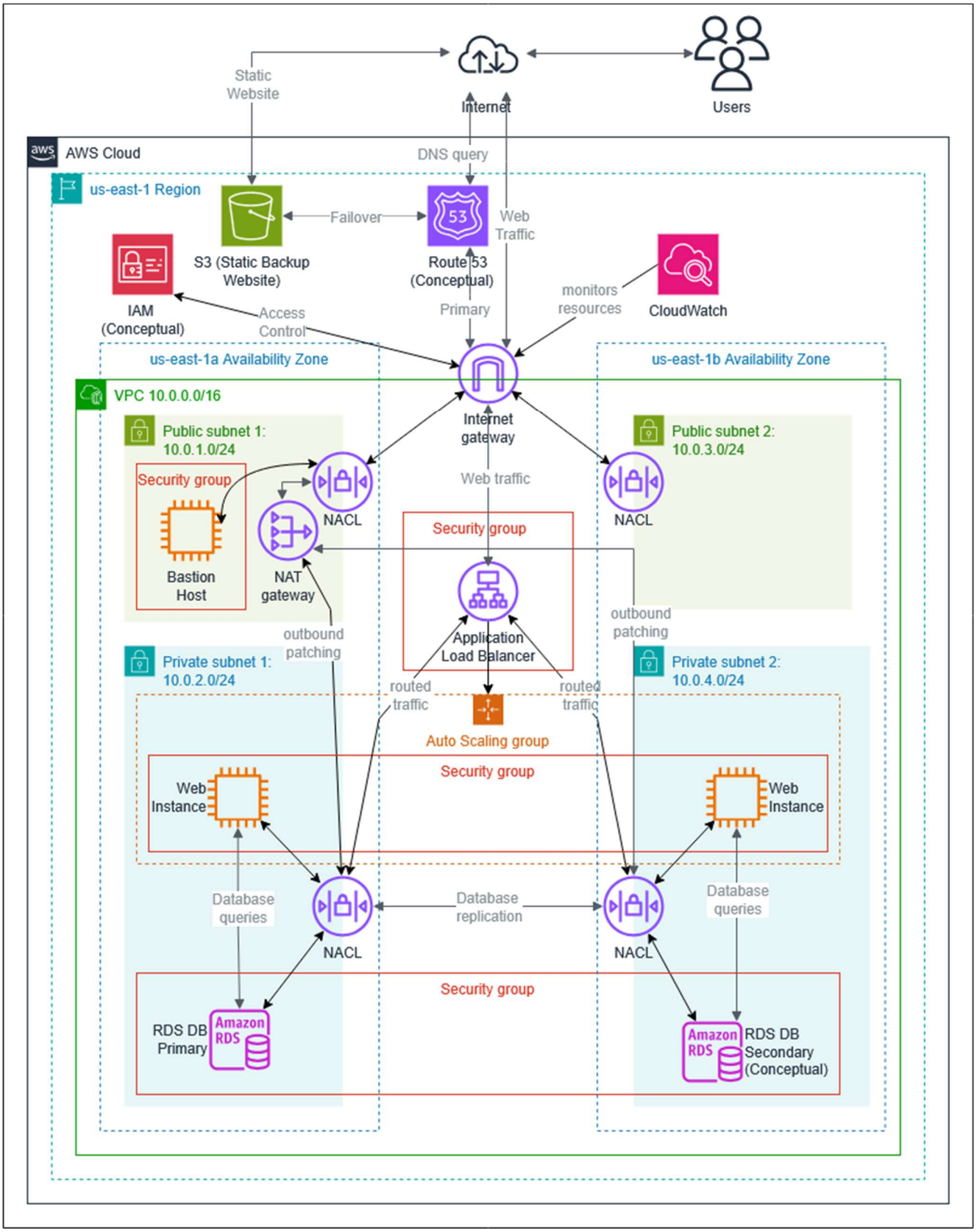
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Introduction

WinLocal Giveaways is a growing competition platform that currently operates on an on-premises infrastructure consisting of a NetScaler load balancer, web and application servers, a Microsoft SQL database server, and a NAS storage device. The existing setup has caused several problems that are limiting the company's growth, including an outdated and unsupported load balancer causing periodic outages, a database server that frequently runs out of disk space, and a flat network architecture that contributed to a ransomware attack in which attackers remained undetected for approximately ten weeks while exfiltrating sensitive customer data.

This report documents the design and implementation of a cloud-based Proof of Concept solution for WinLocal Giveaways on Amazon Web Services. The solution migrates the company's competition platform to a secure, scalable and highly available AWS architecture deployed in the us-east-1 region. Task 1 presents the full proposed architecture as a draw.io diagram including conceptual components beyond the PoC. Task 2 details the implementation of the core PoC infrastructure including the VPC, EC2 instances, RDS database, Application Load Balancer, Auto Scaling Group, and S3 static backup website, along with screenshots and explanations of each resource. Task 3 justifies two additional proposed AWS services beyond the PoC, namely IAM and Route 53, that would add further security and resilience value to the solution. Task 4 evaluates the architecture against the AWS Well-Architected Framework's Sustainability Pillar and provides recommendations for future improvements.

TASK 1: Detailed AWS Architecture Diagram from draw.io (Design Phase)



TASK 2: Proof of Concept (PoC) Implementation and explanation

Networking Design

To ensure the principle of Defence in Depth, this networking design enforces network segmentation by separating public-facing and private resources into different subnets with controlled traffic flows. (Amazon Web Services, 2026c) (Cybersecurity and Infrastructure Security Agency, 2016)

The Virtual Private Network (project-vpc-B00168689) is configured with a CIDR block of 10.0.0.0/16 which provides up to 65,536 private IP addresses. The VPC is deployed in the us-east-1 region across two Availability Zones (us-east-1a and us-east-1b) to ensure high availability. (Amazon Web Services, 2026b)

There are four subnets, two of which are public subnets (10.0.1.0/24 and 10.0.3.0/24) which hosts internet-facing resources such as the Bastion Host, NAT Gateway and Application Load Balancer. The other two subnets are the private subnets (10.0.2.0/24 and 10.0.4.0/24) which hosts the web instances on EC2 and the RDS database, which are never directly accessible from the internet. The two Availability Zones each have one public and one private subnet.

An Internet Gateway (project-vpc-B00168689-igw) is attached to the VPC to allow inbound and outbound internet traffic to the public subnets. The public route table routes all traffic destined for every IPv4 addresses through the Internet Gateway. (Amazon Web Services, 2026c)

A NAT Gateway (project-nat-B00168689) is deployed in Public Subnet 1 to allow the instances in the private subnet to make outbound connections to the internet for patching and updates, without exposing them to inbound internet traffic. Both of the private route tables route all traffic through the NAT Gateway. (Amazon Web Services, 2026d)

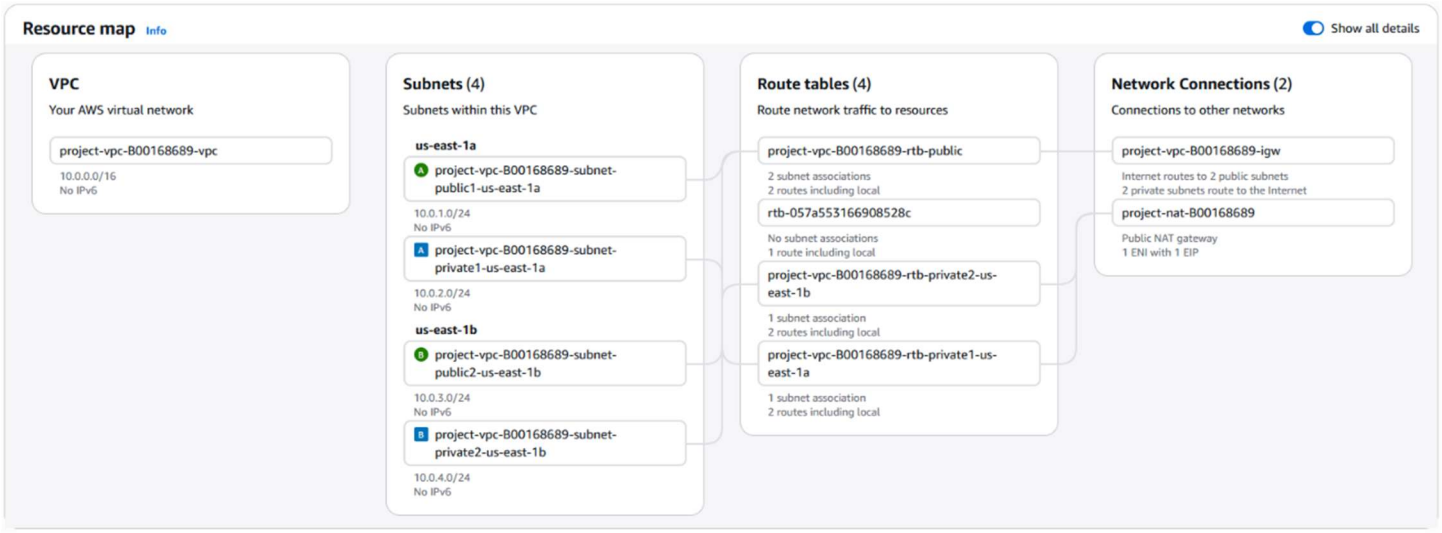


Figure 1 above: VPC resource map for project-vpc-B00168689 showing four subnets, route tables, Internet Gateway and NAT Gateway.

Bastion Host

The Bastion Host (project-bastion-B00168689) is a t3.micro EC2 instance running Amazon Linux 2023 and deployed in Public Subnet 1 in the us-east-1a AZ with its own public IP address. The Bastion Host is necessary because the web instances and RDS database are deployed in private subnets with no public IP addresses as exposing application servers directly to the internet would create significant security risks such as direct exploitation of vulnerabilities in the web application or operating system. The Bastion Host provides a single tightly controlled entry point for administrative access, which is far easier to monitor and audit than multiple individually exposed instances. (Amazon Web Services, 2026e)

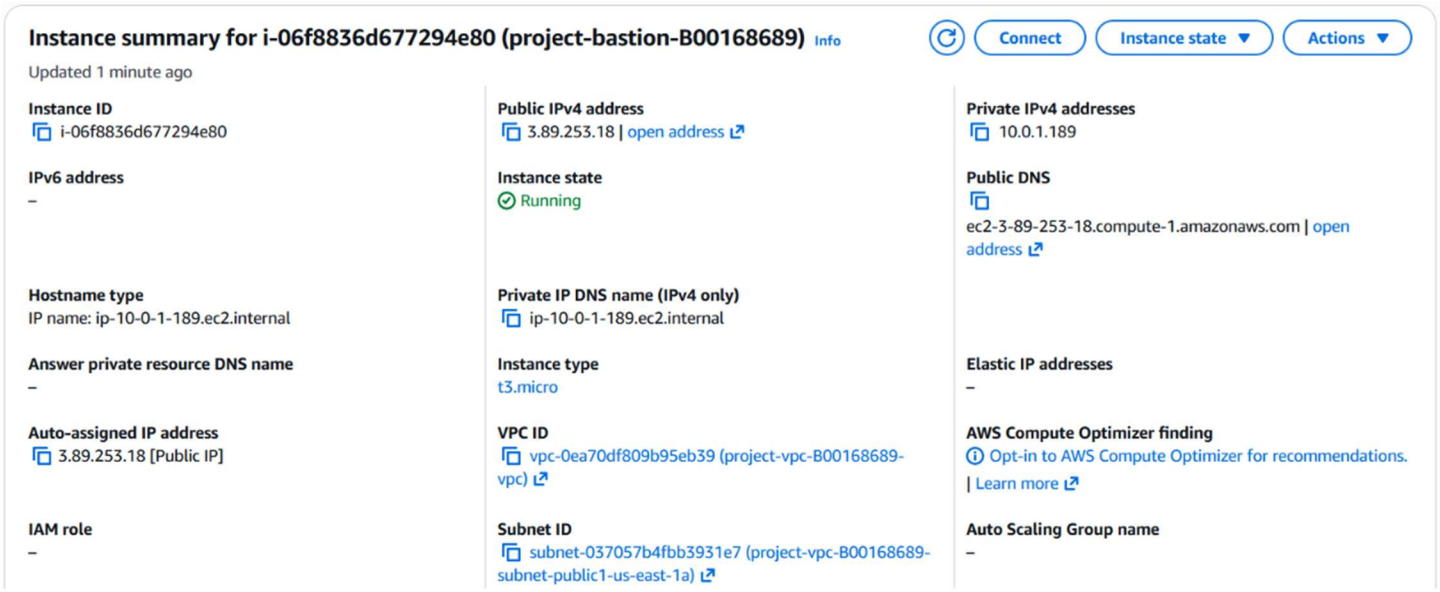


Figure 2 above: project-bastion-B00168689 EC2 instance details showing Running state and Public Subnet 1 deployment.

The Bastion Host has a Security Group (BastionSG) that allows SSH (port 22) access from the administrator's IP. All other inbound traffic is denied by default to prevent unauthorised access.

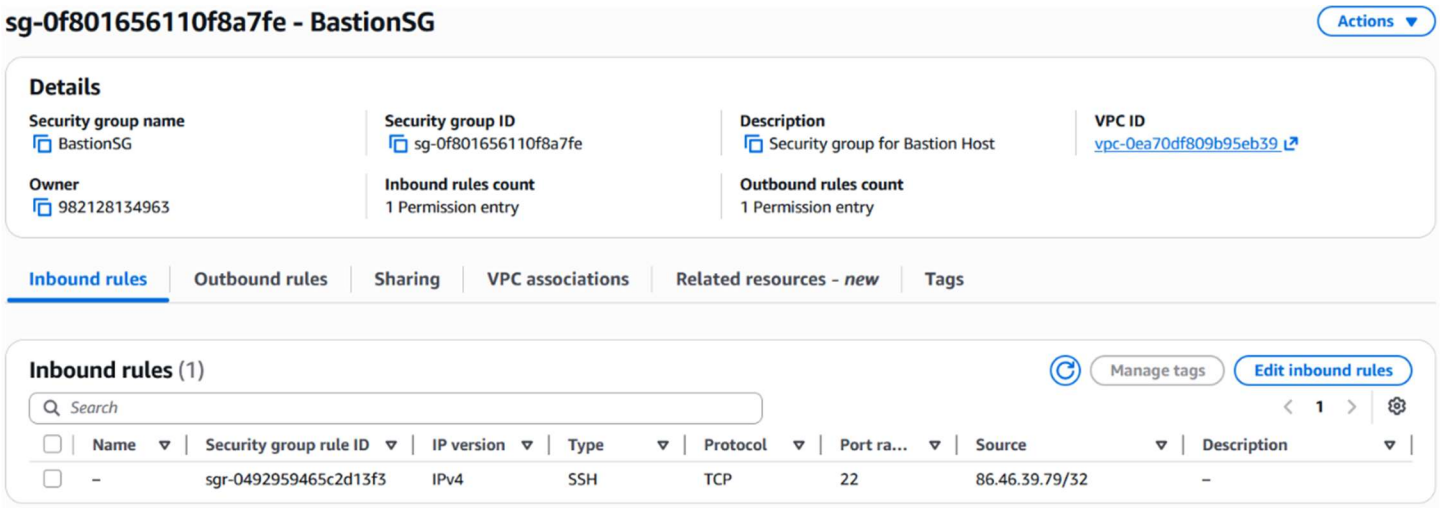


Figure 3 above: BastionSG inbound rules showing SSH (port 22) restricted to the administrator's IP address only.

To access a private web instance, an administrator first opens an SSH connection into the Bastion Host using the .pem key pair, then from the Bastion Host opens another SSH connection into the private instance using the same key. The private instances themselves only allow SSH traffic from BastionSG to make direct SSH connections from the internet to a private instance impossible, even with the key pair. (Amazon Web Services, 2026e)

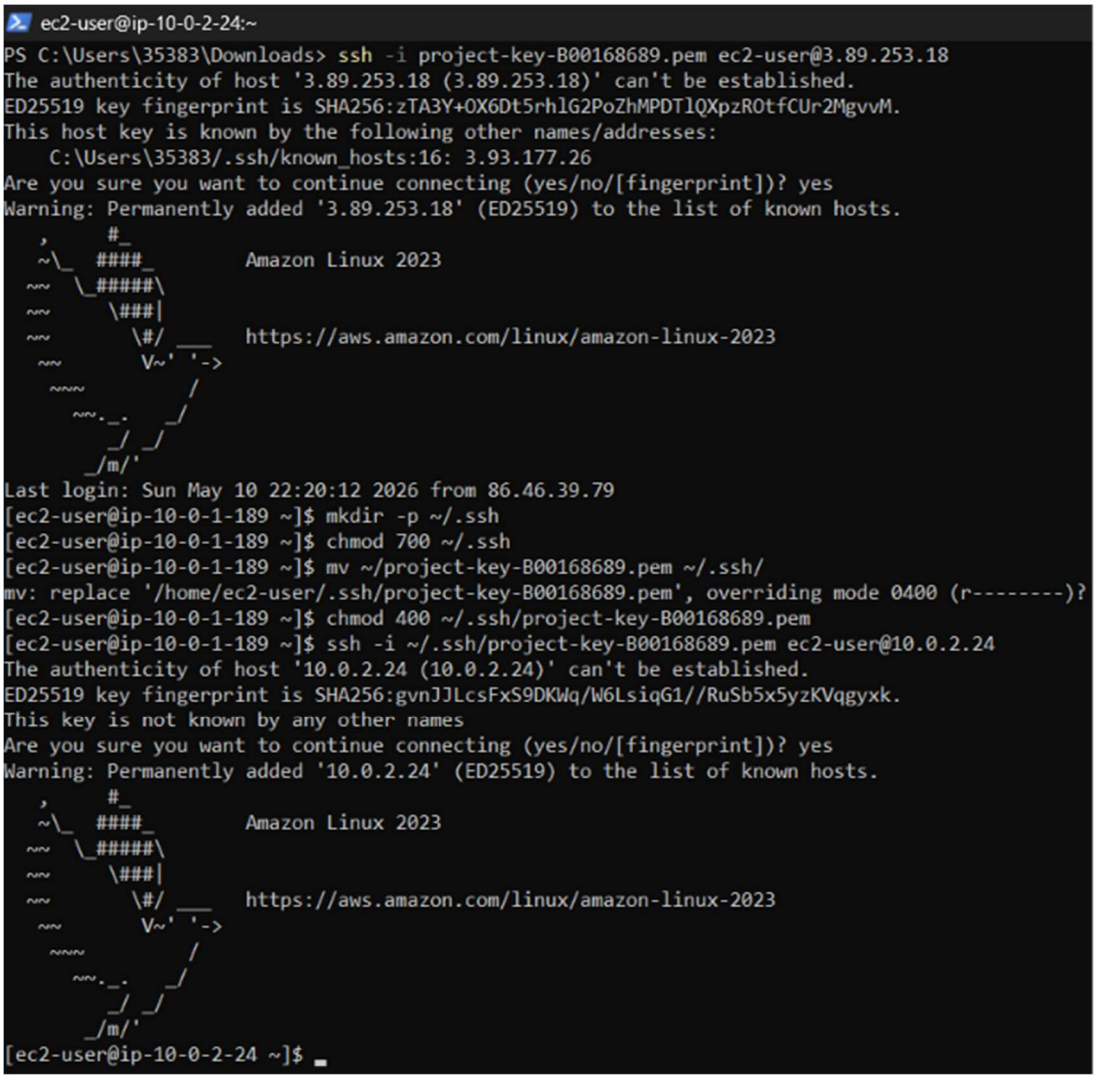


Figure 4 above: PowerShell terminal showing successful SSH connection to the Bastion Host and then into private web instance project-web1-B00168689.

Amazon Machine Image (AMI)

An AMI was created from a running web instance that already had Apache, PHP, the web application and the RDS connection configured. AMIs are useful during competitions as they enable fast and consistent instance launches so that if the Auto Scaling Group needs to scale out a new instance due to high traffic, it can launch from the AMI with everything pre-configured rather than running the User data script from scratch, significantly reducing the time before

the new instance is ready to serve traffic. They can also act as a backup in cases where a web instance becomes corrupted or fails, a replacement can be launched from the AMI in minutes. (Amazon Web Services, 2026a)

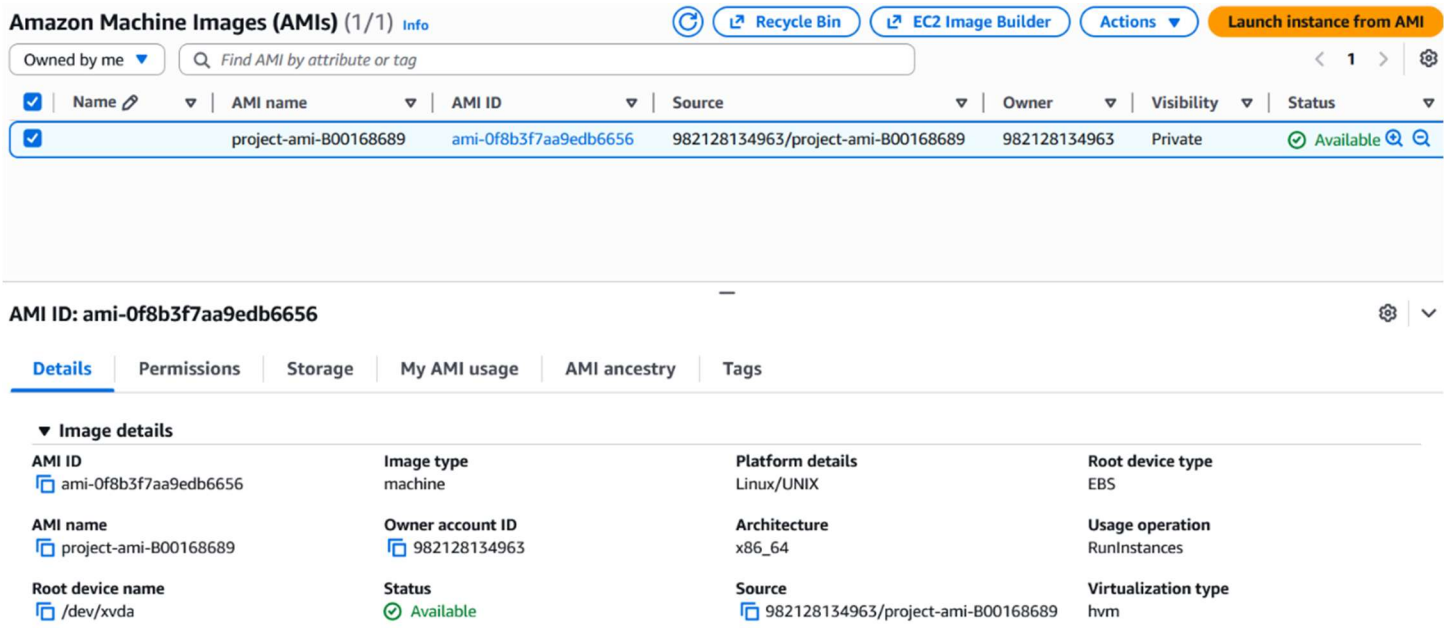


Figure 5 above: project-ami-B00168689 AMI details tab showing Available status and source instance.

Single-AZ Amazon RDS Deployment

The RDS instance (project-db-B00168689) is a MySQL database deployed in a Single-AZ configuration in us-east-1a and hosted in a subnet group containing only the two private subnets. As public access is disabled, the database is never directly reachable from the internet and all connections must come from within the VPC via the PrivateInstanceSG security group on port 3306 which is used for MySQL.

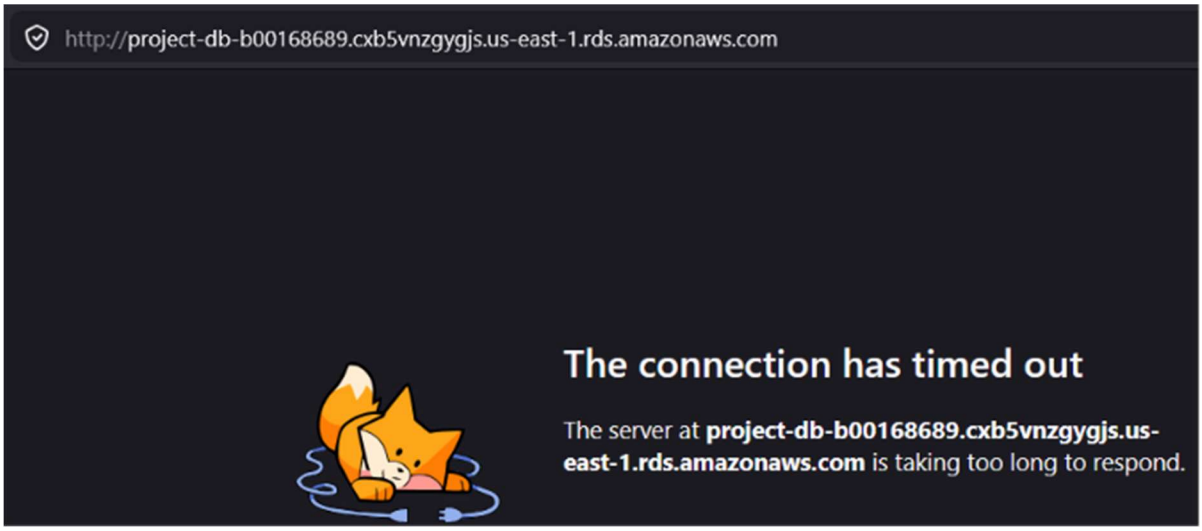


Figure 6 above: connection attempt to project-db-B00168689's endpoint on browser times out.

The web application connects to the RDS database using a configuration file containing the RDS endpoint, database name, username and password, to retrieve and display contact records from the database.

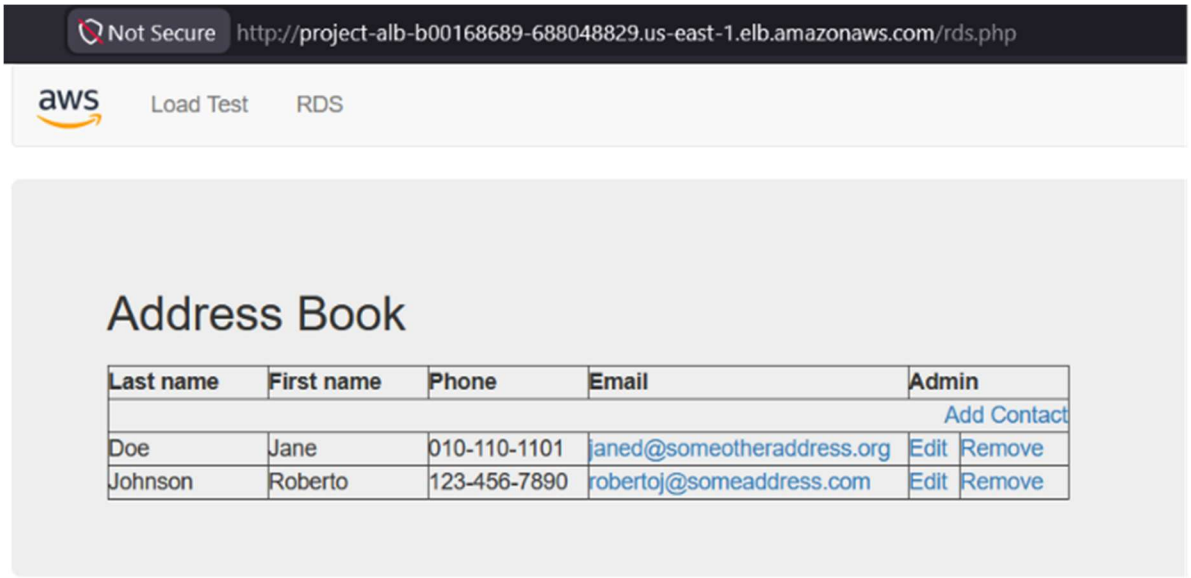


Figure 7 above: Address Book web application showing contact records retrieved from the RDS database via the ALB DNS address:

The Single-AZ deployment means if the Availability Zone the RDS instance is running on experiences an outage, the database becomes completely unavailable and the application loses its data layer entirely meaning that competition entries cannot be submitted or retrieved until the issue is resolved. To improve availability, a Multi-AZ RDS deployment can be implemented which would automatically create a synchronous standby replica in us-east-1b that would take over if there was an outage in us-east-1a. (Amazon Web Services, 2026f)

Auto Scaling Configuration

Firstly, a launch template (project-lt-B00168689) was created to define the configuration for instances launched by the Auto Scaling Group, including the AMI (project-ami-B00168689), instance type, security group and user data script. As the AMI was created from a running web instance with Apache, PHP. and the web application already installed, new instances will launch quickly with all the necessary configurations.

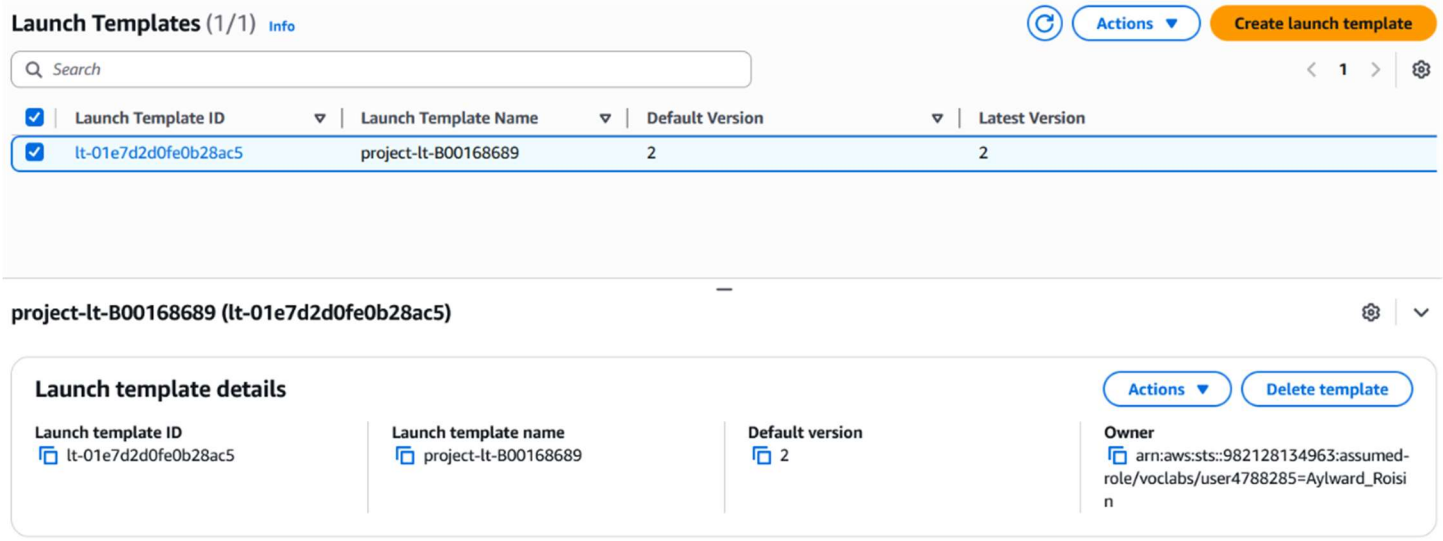


Figure 8 above: project-lt-B00168689 launch template details showing AMI, instance type, and security group configuration.

The Auto Scaling Group (project-asg-B00168689) was then configured with both the desired capacity and minimum capacity set to one instance for the website to run on as it is believed to be sufficient for handling the expected level of traffic most of the time. The maximum capacity was set to two so that it can scale out to an extra instance automatically when demand increases during competitions. Additionally, the AGS will automatically launch a replacement if an instance becomes unhealthy or terminates unexpectedly to ensure the web application is available without any manual intervention. (Amazon Web Services, 2026g)

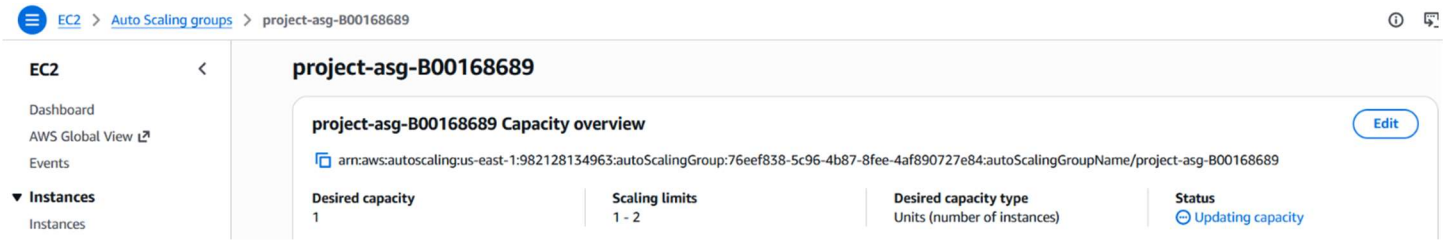


Figure 9 above: project-asg-B00168689 Auto Scaling Group showing the Desired capacity, Minimum capacity, and Maximum capacity.

A dynamic scaling policy (project-scaling-policy-B00168689) was also configured to maintain the average CPU utilisation at 60% to set the threshold for when an instance should be added or removed. Whenever the CloudWatch monitoring tool notices the CPU load across the running instance exceeding the 60%, the ASG automatically launches a second instance to distribute the load. Likewise, when CPU utilisation drops back below 60%, the ASG will scale back in and terminate the second instance after a cooldown period of five minutes to ensure it is not a brief dip. (Amazon Web Services, 2026h)

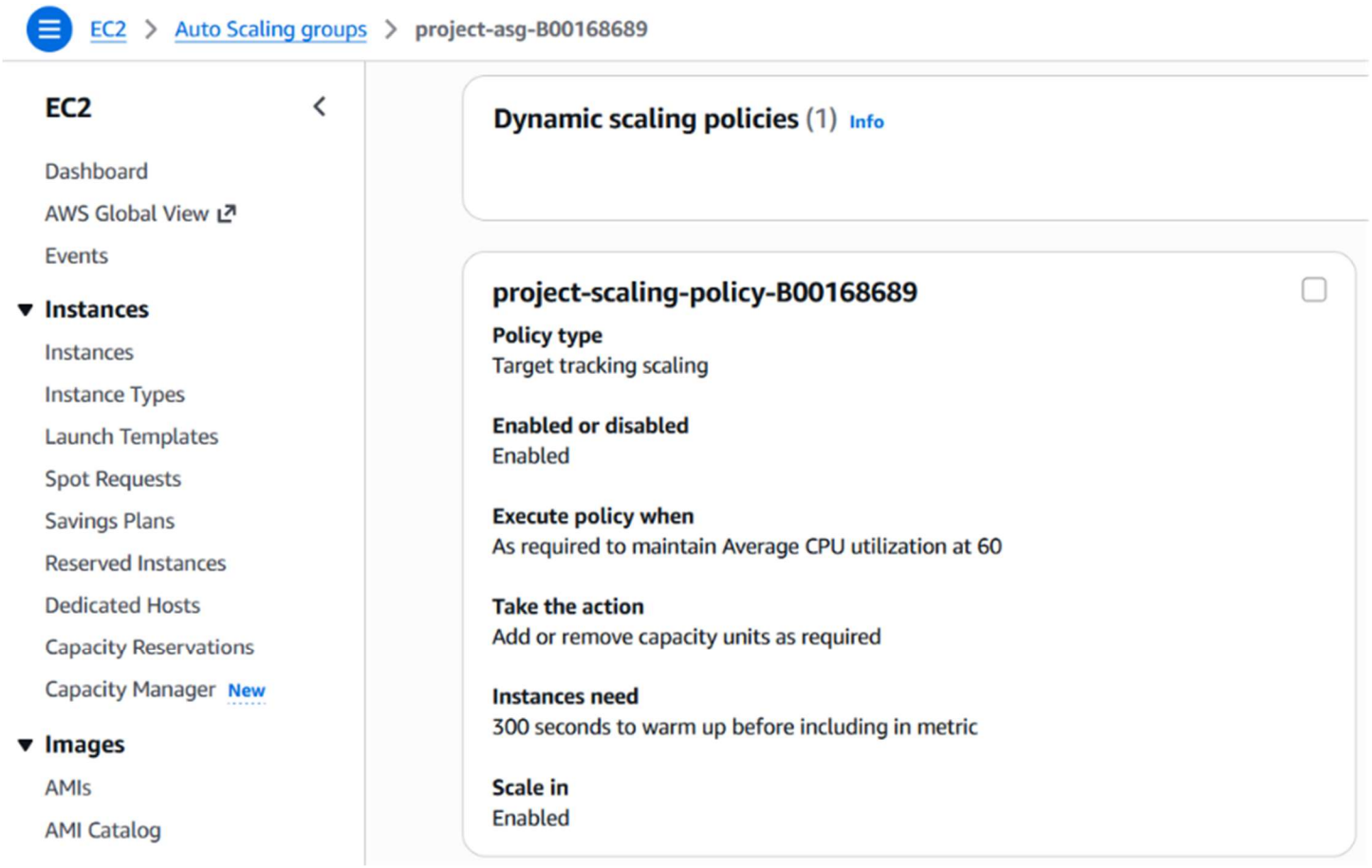


Figure 10 above: project-scaling-policy-B00168689 target tracking scaling policy configured to maintain 60% average CPU utilisation.

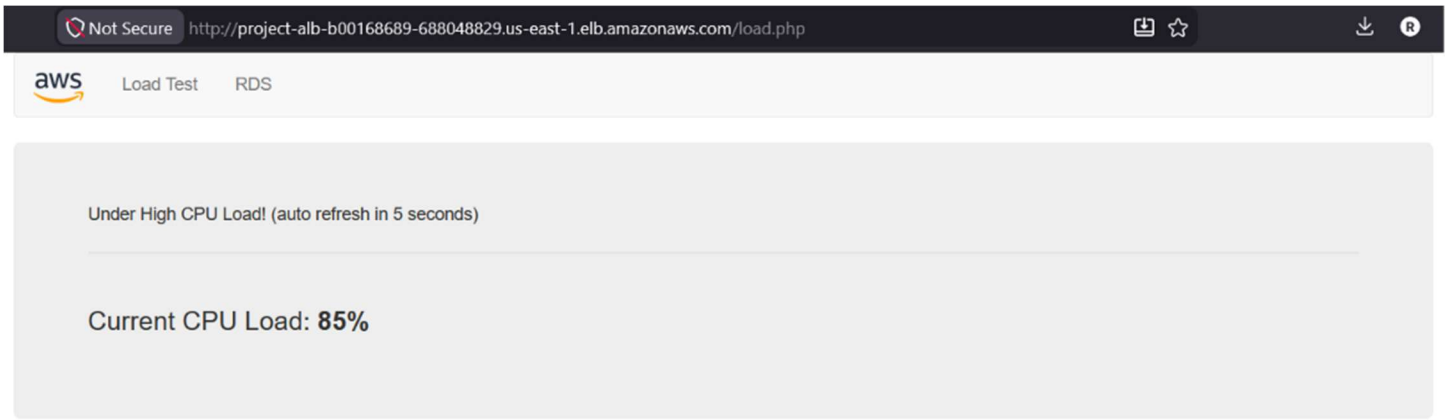


Figure 11 above: load test triggering CPU above 60%.

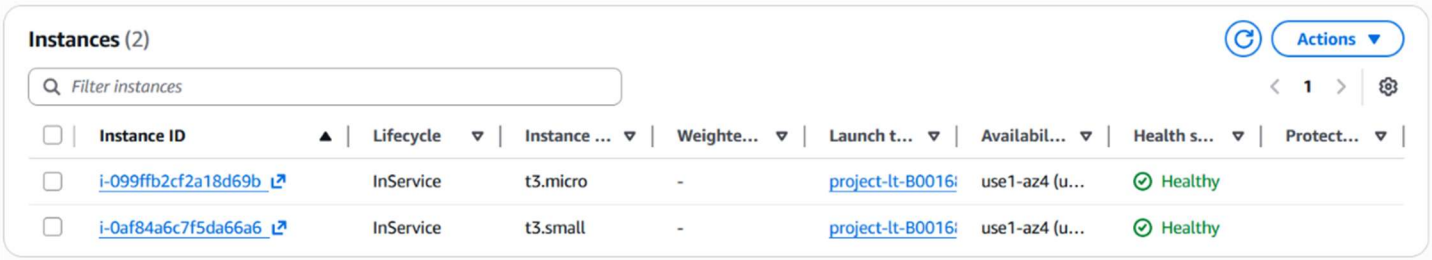


Figure 12 above: the Auto Scaling Group launches a second instance in response.

Load Balancer configuration

An Application Load Balancer (project-alb-b00168689) was deployed across both of the public subnets in both the us-east-1a and us-east-1b Availability Zones. New instances launched by Auto Scaling Group during scale-out events are automatically registered to ALB to begin receiving traffic immediately. It was chosen as the load balancer type as ALB is specifically designed to receive incoming HTTP traffic from the internet and distribute it across the web instances in the private subnets evenly to ensure neither is overwhelmed. ALB also performs health checks on the web instances and automatically stops routing traffic to any instance that becomes unhealthy so that users are never directed to a failed server. (Amazon Web Services, 2026i)

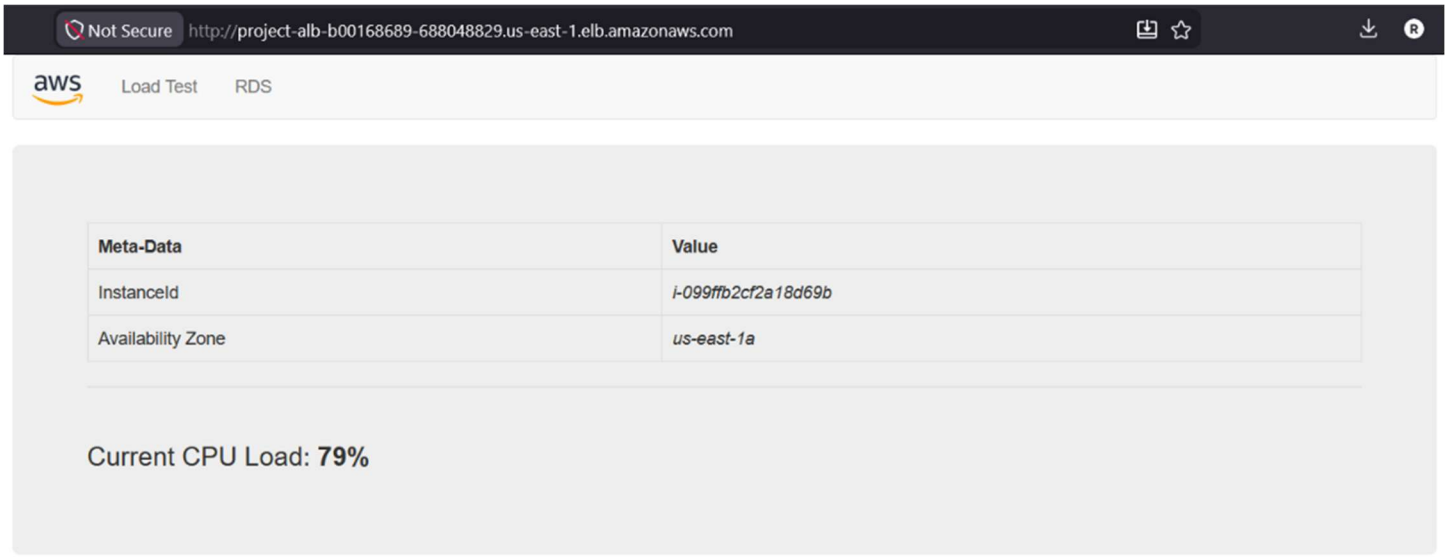


Figure 13 above: ALB DNS address returning instance ID from us-east-1a confirming traffic distribution.

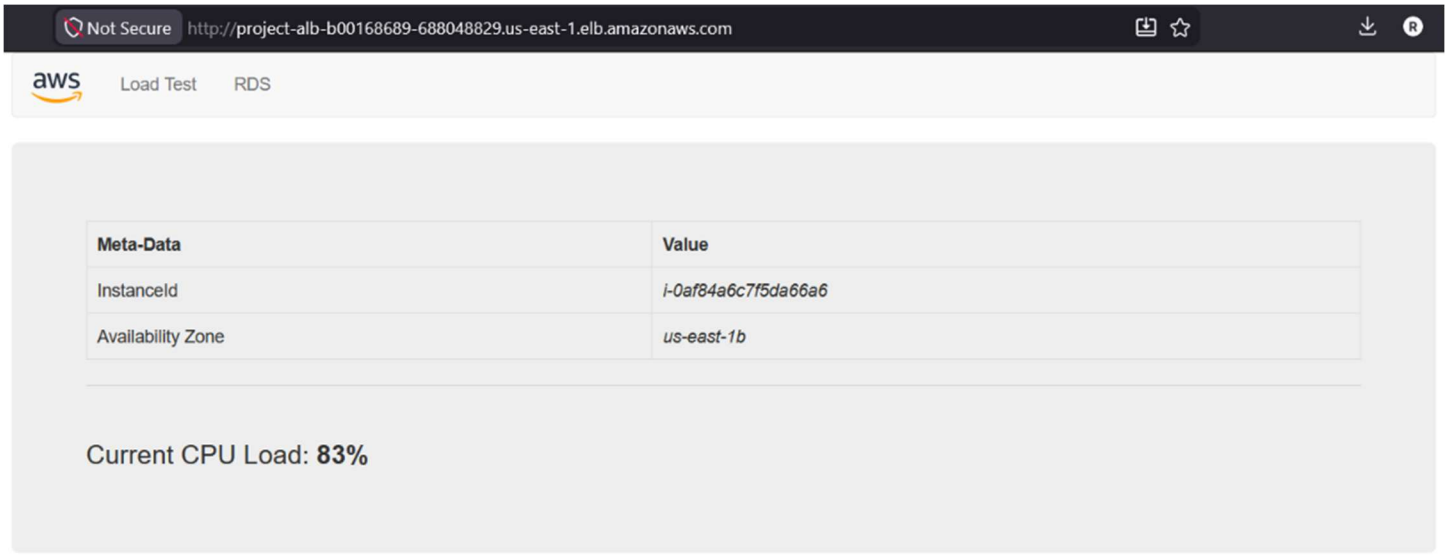


Figure 14 above: ALB DNS address returning instance ID from us-east-1b confirming traffic distribution across both instances.

Security Implementation

Four Security Groups were configured for each instance types to act as a stateful firewalls that follow the Principle of Least Privilege where each resource only allows the specific traffic it needs. BastionSG only allows inbound SSH (port 22) from the administrator's IP address. ALBSG allows inbound HTTP (port 80) and HTTPS (port 443) from anywhere (0.0.0.0/0) into the load balancer. The EC2 web instances' PrivateInstanceSG only allows inbound SSH from BastionSG and inbound HTTP/HTTPS from ALBSG. RDSSG only allows inbound MySQL (port 3306) from PrivateInstanceSG into the database. All of the Security Group allow any outbound traffic to any destination to permit return traffic which can be sent in a different (ephemeral) port. (Amazon Web Services, 2026j)

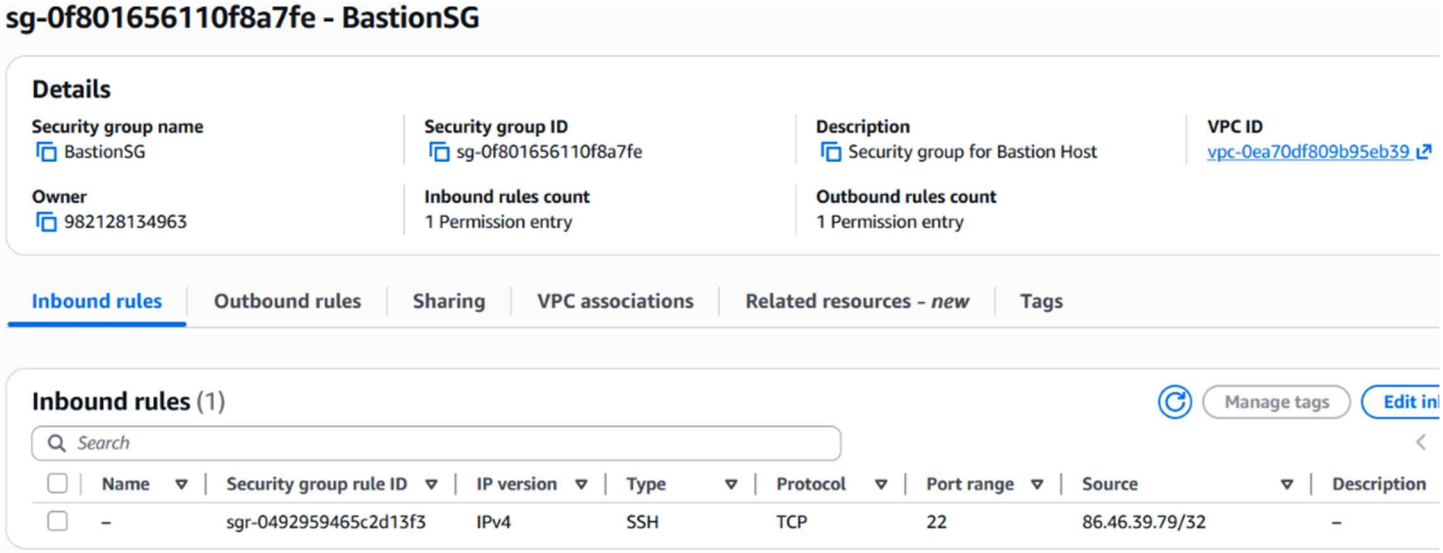


Figure 15 above: BastionSG inbound rules showing SSH (port 22) restricted to administrator IP only.

sg-0ea20f71c057217bc - ALBSG

Details

Security group name	Security group ID	Description	VPC ID
ALBSG	sg-0ea20f71c057217bc	Application Load Balancer	vpc-0ea70df809b95eb39

Owner	Inbound rules count	Outbound rules count
982128134963	1 Permission entry	1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (1)

Manage tags | Edit in

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
–	sgr-06608c5776b78bf2e	IPv4	HTTP	TCP	80	0.0.0.0/0	–

Figure 16 above: PrivateInstanceSG inbound rules showing SSH from BastionSG and HTTP/HTTPS from ALBSG only.

sg-0dfb065d140782b72 - PrivateInstanceSG

Details

Security group name	Security group ID	Description	VPC ID
PrivateInstanceSG	sg-0dfb065d140782b72	Security group for private web instances	vpc-0ea70df809b95eb39

Owner	Inbound rules count	Outbound rules count
982128134963	2 Permission entries	1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (2)

Manage tags | Edit inl

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Des
–	sgr-0a0e49dcd2c57ce18	–	SSH	TCP	22	sg-0f801656110f8a7fe / BastionSG	–
–	sgr-09cc4d7ac703815d1	–	HTTP	TCP	80	sg-0ea20f71c057217bc / ALBSG	–

Figure 17 above: RDSSG inbound rules showing MySQL (port 3306) from PrivateInstanceSG only.

sg-03b1ff9f34655efa7 - RDSSG

Details

Security group name	Security group ID	Description	VPC ID
RDSSG	sg-03b1ff9f34655efa7	Created by RDS management console	vpc-0ea70df809b95eb39

Owner	Inbound rules count	Outbound rules count
982128134963	2 Permission entries	1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (2)

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Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
–	sgr-06b538e5578be0ccd	–	MYSQL/Aurora	TCP	3306	sg-0dfb065d140782b72 / PrivateInstanceSG
–	sgr-081cc5fb50b2145c9	IPv4	MYSQL/Aurora	TCP	3306	86.46.39.79/32

Figure 18 above: ALBSG inbound rules showing HTTP (port 80) and HTTPS (port 443) from 0.0.0.0/0.

Four custom NACLs were configured for each subnet to provide an additional layer of traffic filtering at the subnet boundary. Both inbound and outbound rules were explicitly defined as NACLs are stateless. Public Subnet 2's NACL only allows inbound HTTP, HTTPS, and ephemeral ports (1024-65535) to permit response traffic back from the internet, while blocking everything else. Public Subnet 1's NACL has the same inbound rules in addition to allowing SSH due to the Bastion Host. Both private subnet NACLs restrict inbound traffic to within the VPC's CIDR range of 10.0.0.0/16 for HTTP, HTTPS, and SSH, with ephemeral ports also allowed from any source to permit response traffic from outbound requests made through the NAT Gateway. All four NACLs allow all outbound traffic to ensure responses can reach their destinations. (Amazon Web Services, 2026k) (Amazon Web Services, 2026l)

acl-09a720ecb052e87fd / NACL-Public1-B00168689

Actions

Details Info

Network ACL ID	Associated with	Default	VPC ID
acl-09a720ecb052e87fd	subnet-037057b4fbb3931e7 / project-vpc-B00168689-subnet-public1-us-east-1a	No	vpc-0ea70df809b95eb39 / project-vpc-B00168689-vpc

Owner
982128134963

Inbound rules | Outbound rules | Subnet associations | Tags

Inbound rules (5)

Edit inbound rules

Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	SSH (22)	TCP (6)	22	86.46.39.79/32	Allow
110	HTTP (80)	TCP (6)	80	0.0.0.0/0	Allow
120	HTTPS (443)	TCP (6)	443	0.0.0.0/0	Allow
130	Custom TCP	TCP (6)	1024 - 65535	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Figure 19 above: NACL-Public1-B00168689 inbound and outbound rules

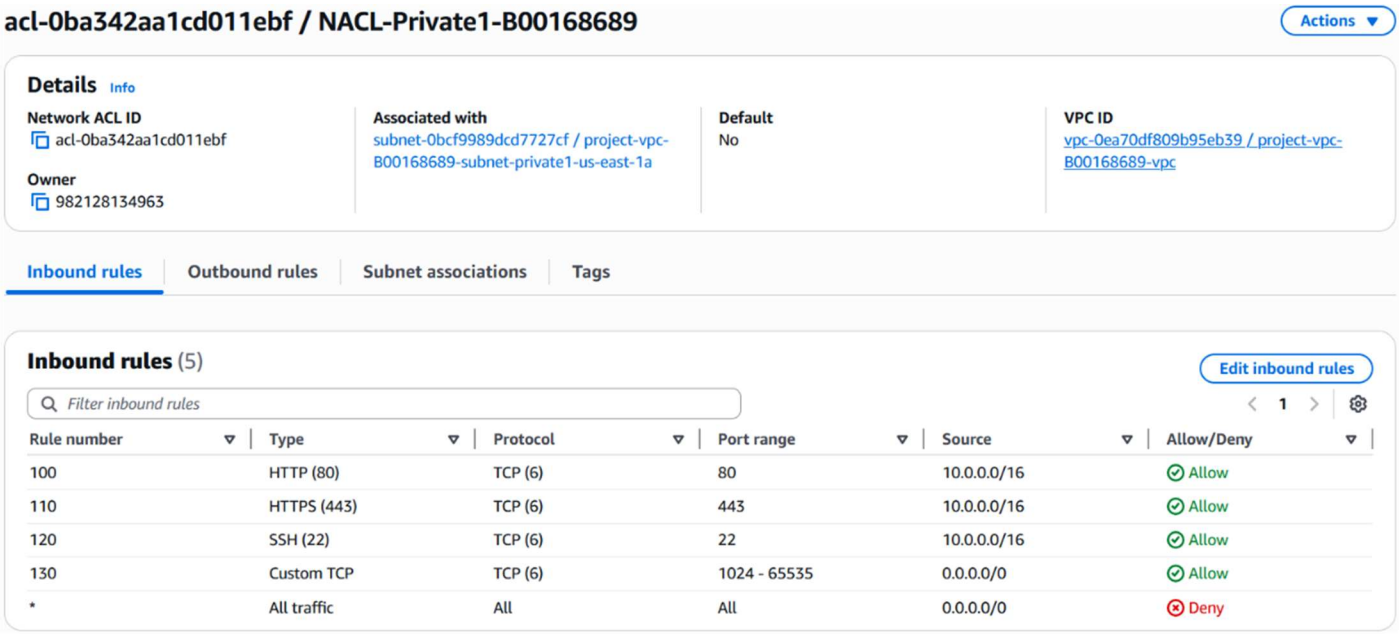


Figure 20 above: NACL-Private1-B00168689 inbound and outbound rules.

Static Backup Website

A static backup website was created and hosted on Amazon S3 (project-backup-b00168689) in situations where WinLocal's primary Giveaways application becomes unavailable during an Availability Zone outage, a failed deployment, or an unexpected surge in traffic that overwhelms the web instances before Auto Scaling can respond. S3 is fully managed by Amazon and it costs almost nothing to host a simple static HTML page (Amazon Web Services, 2026n). It is considered the most durable cloud storage with 11 nines of data durability. S3 has four nines of availability which is more reliable than the primary application which depends on the EC2 instances, RDS and ALB all being healthy simultaneously. (Amazon Web Services, 2026m)

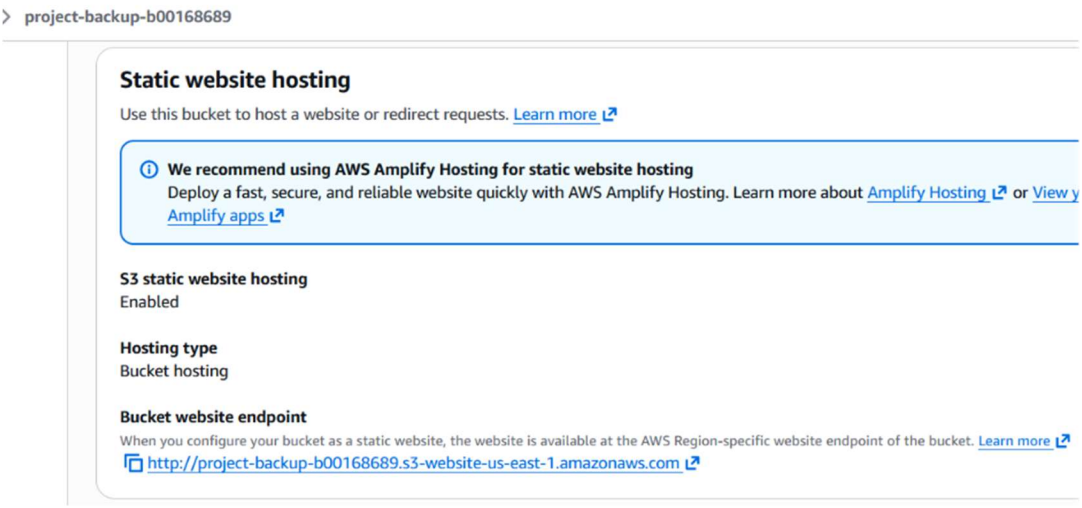


Figure 21 above: static website hosting enabled.

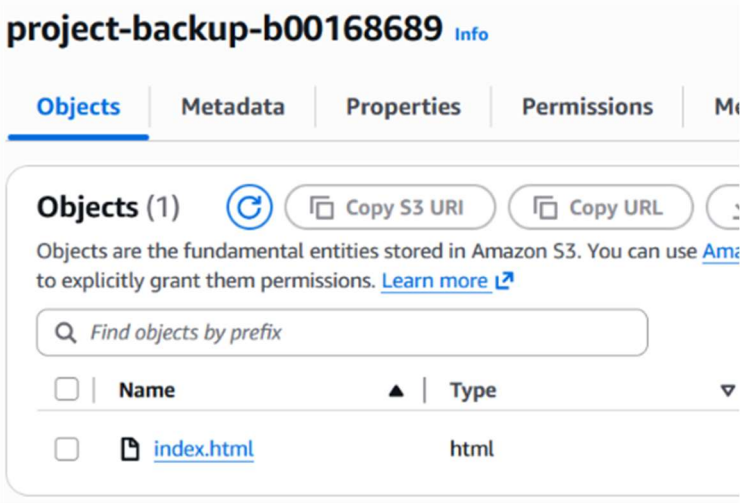


Figure 22 above: bucket policy was configured to allow public read access to the index.html using the S3 website endpoint URL.

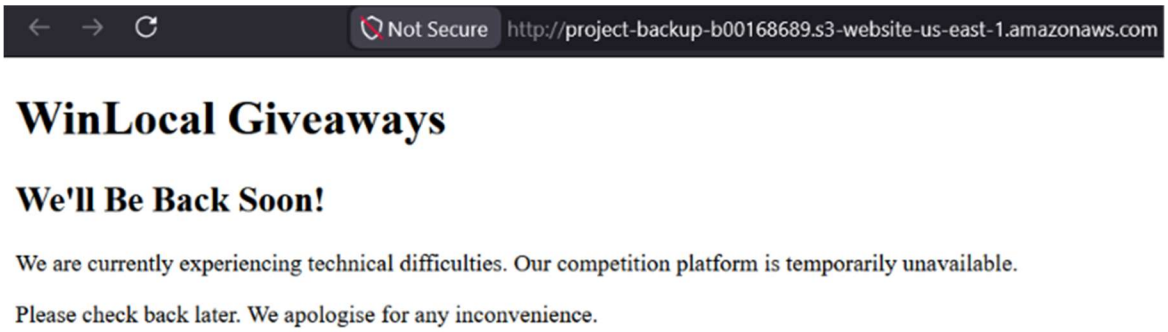


Figure 23 above: WinLocal Giveaways static backup website loading in browser via S3 website endpoint URL.

TASK 3: Justification of Additional AWS Services (Beyond PoC)

Identity and Access Management

AWS Identity and Access Management (IAM) is a service that allows organisations to control who can access specific services and resources and what actions they are permitted to perform on those resources. IAM users and roles have no permissions by default and access must be explicitly granted through IAM policies rather than restricted after the fact (Amazon Web Services, 2026p). IAM is free to use, meeting WinLocal's cost optimisation requirement as a relatively new organisation with a small budget (Amazon Web Services, 2026q).

IAM is relevant to WinLocal as the investigation into the recent ransomware attack suggested that poor access control practices contributed significantly to the breach, with staff susceptibility to phishing and weak password practices allowing attackers to gain and retain access without triggering any alarms. This directly highlights the absence of a formal and enforced access control policy, which IAM is specifically designed to address.

In addition to the threat of a cyberattack, WinLocal is also vulnerable to a data breach caused internally as the company collects personally identifiable information from competition entrants such as names and addresses. The General Data Protection Regulation (GDPR) requires organisations in operating in the European Union to implement technical measures to protect personal data, and overly permissive IAM access means a single compromised credential could expose the entire dataset and lead to the organisation being heavily fined (European Union, 2016).

Implementation of IAM would follow the Principle of Least Privilege which states that users should only be granted the permissions required to perform their tasks, balancing usability, efficiency, and security (Amazon Web Services, 2026o). In practice for WinLocal, the developer would only be granted permissions to deploy and update application code, the two system administrators would have distinct and restricted permissions for infrastructure management to ensure separation of duties, and accounts/finance staff would only have access to AWS Billing and Cost Management console to monitor and report on AWS spending without being able to interact with the infrastructure itself. Additionally, Marketing and customer engagement staff would have read-only access to the specific S3 bucket containing campaign assets or competition entry reports, and to CloudWatch dashboards to view traffic metrics and request counts. To allow the interns to learn by observing infrastructure without being able to modify anything, an assigned IAM identity would enable them to have read-only access to EC2, S3, and CloudWatch dashboards with no write permissions on any resource or access to IAM or AWS Billing. No role would have blanket administrator access that could potentially be abused. (Amazon Web Services, 2026o)

Beyond access control, IAM also supports Multi-Factor Authentication (MFA), which would directly address the phishing vulnerability identified in the investigation regarding the ransomware. Even if a staff member's password was compromised through a phishing attack, MFA would prevent an attacker from using those credentials to access the AWS environment without also having physical access to the MFA device. (Amazon Web Services, 2026p)

Route 53

Route 53 is AWS's scalable DNS service which translates domain names into the IP addresses that browsers use to connect to servers (Amazon Web Services, 2026r). Beyond basic DNS resolution, Route 53 also supports health checks and routing policies, addressing WinLocal's website availability issues as an automatic DNS-based failover mechanism that redirects users to a static backup webpage hosted on Amazon S3 when the primary application becomes unavailable (Amazon Web Services, 2026s). The current Proof of Concept implements the ALB and the S3 static backup website, but without Route 53 to connect them, the failover cannot happen automatically.

Route 53 is relevant to WinLocal as their existing infrastructure uses an outdated and unsupported load balancer that causes periodic failures, resulting in the website becoming completely unavailable. A competition platform that goes down during a promotional campaign does direct damage to WinLocal's reputation and results in lost entries.

Implementation of Route 53 would involve configuring a failover routing policy with two DNS records. The primary record would point to the Application Load Balancer serving the live competition platform, and the secondary record would point to the S3 static backup website. Route 53 health checks would continuously monitor the ALB endpoint and would automatically begin responding to DNS queries using the secondary record instead of the primary one if it became unreachable. Users trying to access the WinLocal website during an outage would be seamlessly redirected to the S3 backup page displaying a competition unavailable message rather than encountering a browser error, with no manual intervention required from the small technical team. (Amazon Web Services, 2026s)

Without Route 53, the S3 static backup website exists but users have no way of being automatically redirected to it during an outage. They would need to manually type in the S3 endpoint URL, which is unrealistic for a general competition audience. Route 53 makes the failover invisible to the user, which is the level of resilience WinLocal needs.

Route 53 also provides high availability as a globally distributed DNS service with no single point of failure. It also integrates with AWS Certificate Manager to enable HTTPS for a custom domain name, which would be important for protecting the personal data submitted through the competition entry form and for maintaining user trust given. From a cost perspective, Route 53 charges per hosted zone and per million DNS queries, meaning the cost for a startup with WinLocal's traffic volumes would be minimal and easily justified by the availability improvement it delivers. (Amazon Web Services, 2026r)

TASK 4: Sustainability (AWS Well-Architected Framework)

The Sustainability Pillar of the AWS Well-Architected Framework focuses on minimising the environmental impact of cloud workloads through efficient resource usage reduced waste, and the adoption of managed services (Amazon Web Services, 2026x). Environmental sustainability is a shared responsibility between customers and AWS (Amazon Web Services, 2026u). The WinLocal Giveaways' Proof of Concept architecture has made several decisions that align with the sustainability pillar, and there are further improvements that could be made as the platform grows.

The most impactful contribution to sustainability in the PoC is the Auto Scaling Group. The AWS Well-Architected sustainability pillar recommends maximising utilisation by right-sizing workloads, noting that two hosts running at 30% utilisation are less efficient than one host running at 60% due to baseline power consumption per host (Amazon Web Services, 2026t). The Auto Scaling Group addresses this directly by running a single instance during normal traffic periods and only scaling out to a second instance when CPU utilisation exceeds 60%. This means the architecture avoids running two instances continuously when one is sufficient, and when demand drops back below the threshold, the second instance is automatically terminated after a cooldown period of five minutes (Amazon Web Services, 2026h). WinLocal only consumes the compute resources it actually needs at any given time.

The use of Amazon S3 for the static backup website also contributes to sustainability. Rather than running a dedicated standby EC2 instance to serve the backup page, S3 is a fully managed service where AWS handles all underlying infrastructure and serves the same physical resources to many customers simultaneously, which is more efficient than dedicated single-tenant infrastructure. (Amazon Web Services, 2026n)

An improvement WinLocal could make would be implementing S3 Intelligent-Tiering for any data stored in S3 beyond the static backup website. AWS recommends using managed services such as Amazon EC2 Auto Scaling to adjust capacity to meet demand and Amazon S3 Lifecycle configurations to automatically move infrequently accessed data to cold storage, both of which reduce the total energy required to power a workload. As WinLocal's competition entry data grows, ensuring that historical data is automatically moved to lower-energy storage tiers would reduce unnecessary energy consumption from storing infrequently accessed data on high-performance storage. (Amazon Web Services, 2026v)

Another improvement would be using the AWS Customer Carbon Footprint Tool to measure and track WinLocal's actual carbon emissions from their AWS usage over time to enable the company to set measurable sustainability goals and monitor progress as the platform grows. Setting sustainability targets from an early stage is aligned with the AWS Well-Architected Framework's recommendation to establish long-term sustainability goals for each workload. (Amazon Web Services, 2026w)

Conclusion

This report has presented a cloud-based solution for WinLocal Giveaways that directly addresses the four key issues identified in its current infrastructure. The flat network architecture that enabled the ransomware attack has been replaced with a segmented VPC design with public and private subnets, NACLs, Security Groups and a Bastion Host enforcing the Principle of Defence in Depth. The unreliable on-premises load balancer has been replaced with an Application Load Balancer distributing traffic across two Availability Zones. The disk space limitations of the on-premises database have been resolved by migrating to Amazon RDS MySQL, which provides managed and scalable storage. The requirement for manual scaling during traffic spikes has been addressed through an Auto Scaling Group configured to maintain an average CPU utilisation of 60%, scaling out to a second instance automatically during competitions and scaling back in when demand decreases.

The two conceptual services proposed in Task 3 would further strengthen the solution. IAM would address the access control weaknesses that contributed to the ransomware breach, while Route 53 would complete the automatic DNS-based failover mechanism by connecting the ALB to the S3 static backup website. The architecture also aligns with the AWS Well-Architected Framework's Sustainability Pillar through the use of Auto Scaling to avoid idle resource consumption, and S3 as a managed service for the backup website. Future improvements such as implementing S3 Intelligent-Tiering and the AWS Customer Carbon Footprint Tool would further reduce WinLocal's environmental impact as the platform grows.

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